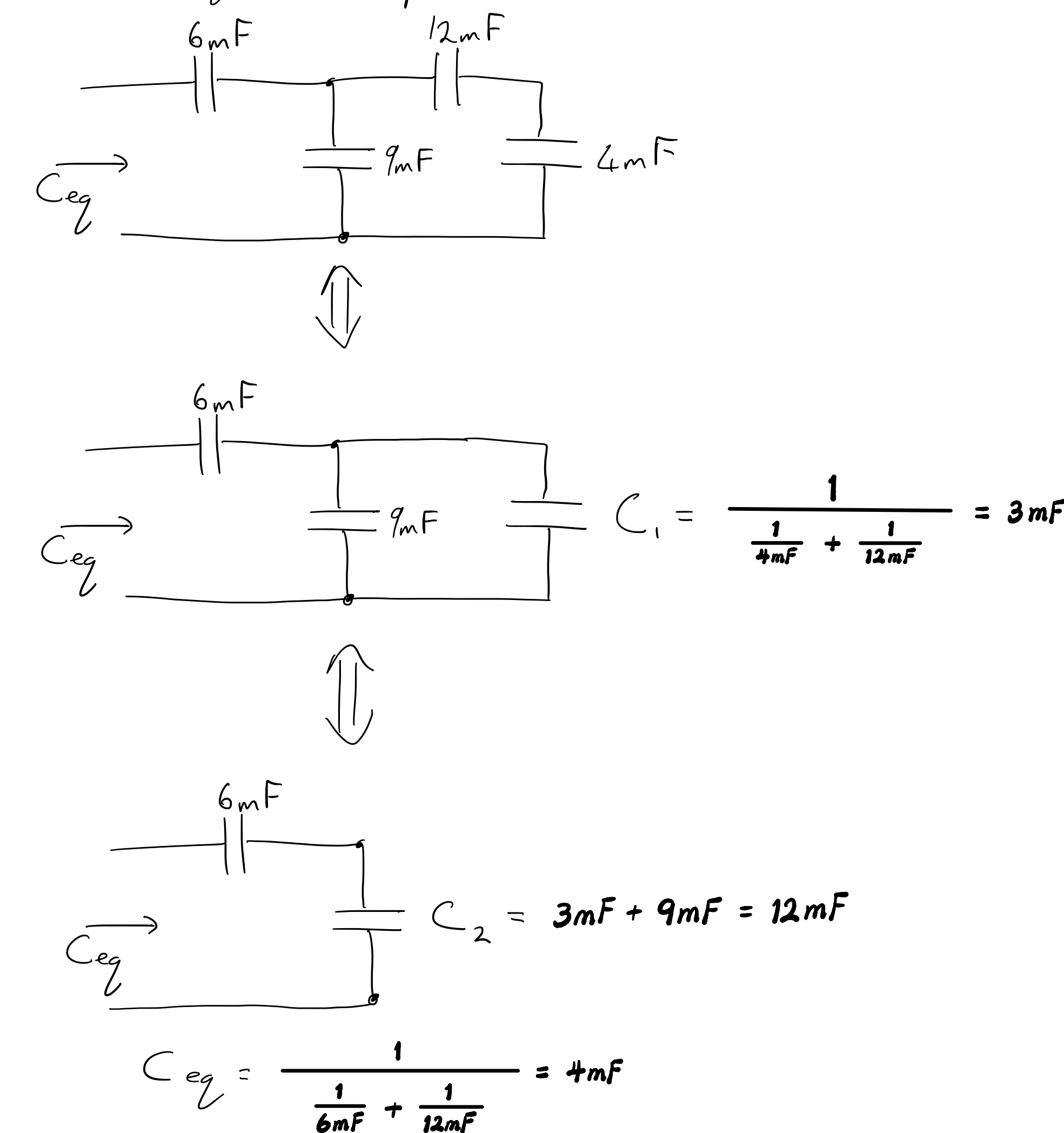


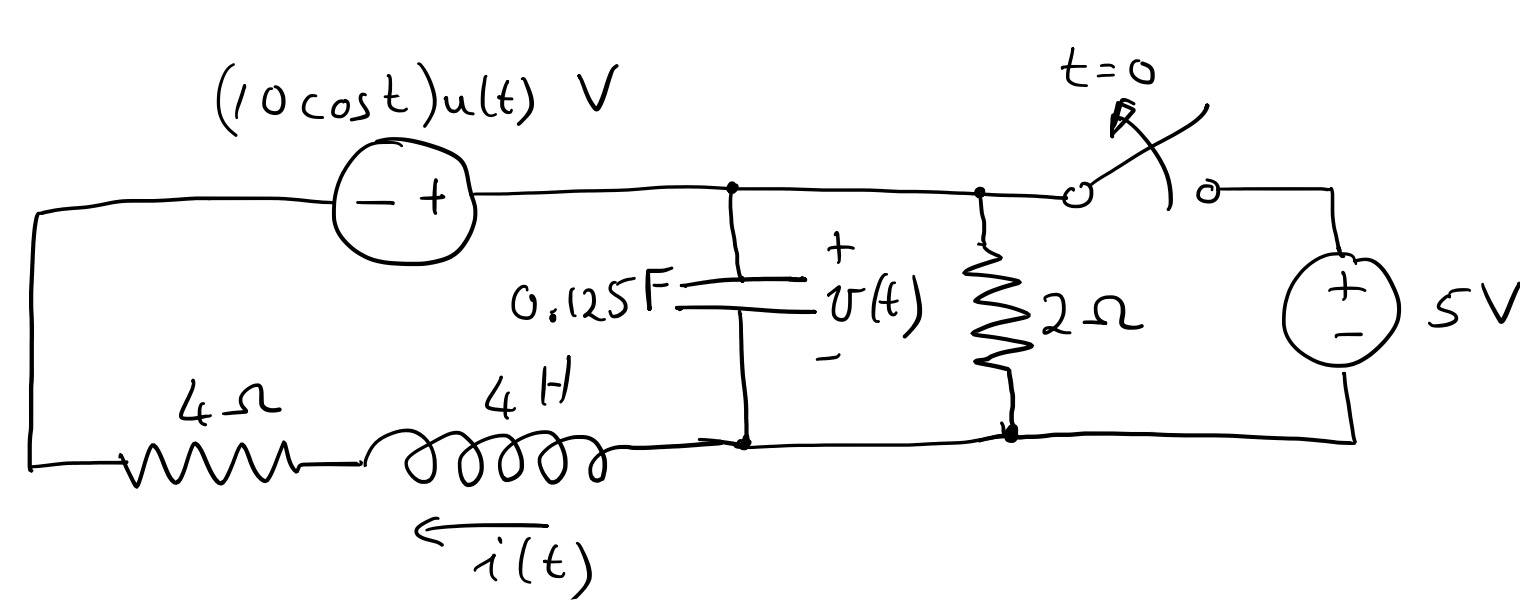
Question 1

Find the equivalent capacitance for the circuit :



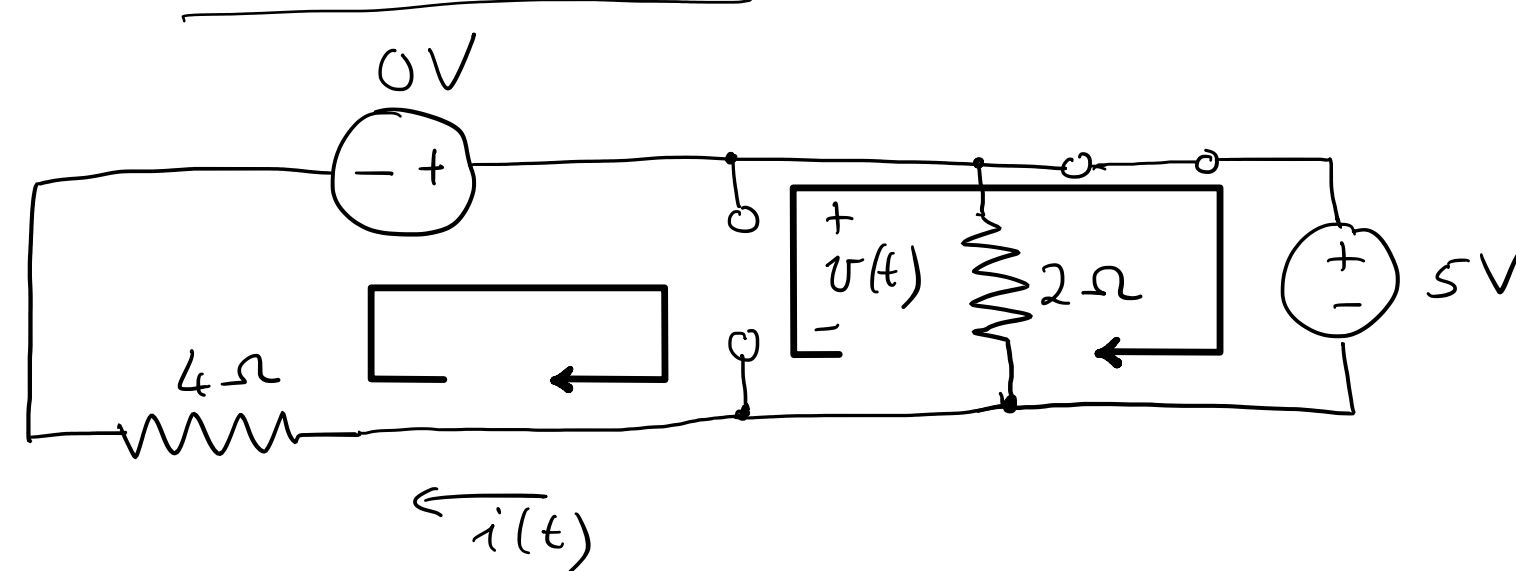
Question 2

A power supply system is modelled by the circuit :



Compute an expression for $v(t)$, $t \geq 0$

- (a) Compute initial conditions
Circuit for $t \leq 0$

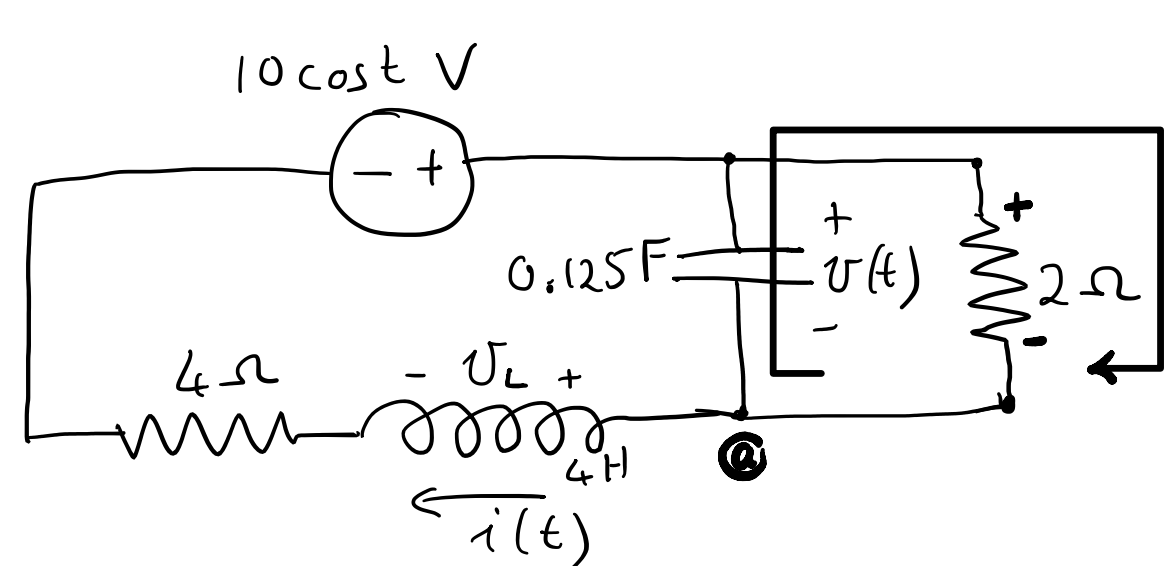


$v(0) = KVL : 5V - v(0) = 0 \Rightarrow v(0) = 5V$

$i(0) = KVL : 0V - v(0) - i(0) \cdot 4\Omega = 0 \Rightarrow i(0) = -\frac{5}{4}A$

- (b) Derive the differential equation for v :

Circuit for $t \geq 0$



KCL at $\odot$: $i = i_C + i_R$ ①

$i_R = \frac{v}{2\Omega} = \frac{1}{2}v(t)$ ②

$i_C = C \frac{dv}{dt} = 0.125 \frac{dv}{dt}$ ③

$\Rightarrow i = \frac{1}{2}v + \frac{1}{8} \frac{dv}{dt}$ ④

KVL around left loop : $10\cos t - v - v_L - 4i = 0$ ⑤

$v_L = L \frac{di}{dt}$ ⑥

Substitute ④ and ⑥ into ⑤ to get

$\frac{dv}{dt} + A \frac{di}{dt} + Bv = v_f$ ⑦

$A = 8 \quad B = 6 \quad v_f = 20\cos t$

Substitute ④ into ⑦ to get

$\frac{d^2v}{dt^2} + C \frac{dv}{dt} + Bv = v_f$ ⑧

$C = 5$

$\therefore \frac{dv}{dt} + 5 \frac{dv}{dt} + 6v = 20\cos t$

- (c) The characteristic equation is : $\lambda^2 + 5\lambda + 6 = 0$

The roots of the characteristic equation are : $\lambda = -2$ and -3

- (d) Recall from part (a) that

$v(0) = 5V$

$i(0) = -\frac{5}{4}A$

and from ④ :

$i(t) = \frac{1}{2}v + \frac{1}{8} \frac{dv}{dt}$

$\Rightarrow$ Initial conditions of ⑧ are :

$v(0) = 5V$ ⑨

$\frac{dv}{dt}(0) = -30V/s$ ⑩

- (e) The solution to ⑧ has the form

1) $v(t) = D e^{-3t} + E e^{-2t}$

② $v(t) = D e^{-3t} + E e^{-2t} + F \sin t + G \cos t$

3) $v(t) = F \sin t + G \cos t$

4) None of the above

The particular solution has to satisfy the ODE ⑧

$\Rightarrow v_p(t) = Ae^{-2t} + Be^{-3t} = 45e^{-2t} - 40e^{-3t}$

The general solution also has to satisfy the ODE ⑧ $v_L(t) = C \cos t + D \sin t = 2 \cos t$

The initial conditions ⑨ and ⑩ could be used to compute values for D and E so that

$\Rightarrow v(t) = 45e^{-2t} - 40e^{-3t} + 2 \cos t$